Missing Plot Analysis

Handling a lost observation in CRD, RCBD and Latin square: direct fitting, Bartlett's covariance method and substitution.

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1 Why plots go missing

Animal or pest damage, flooding, theft, recording errors... Always **record why** an observation is missing.

A poor result **caused by a treatment** is real data and must not be discarded.

A missing plot **destroys the balance** of an RCBD or Latin square; a CRD copes directly.

2 CRD: fit directly

Analyse the observed responses with unequal treatment sample sizes (no substitution needed).

Error df

$$edf = n_{obs} - v$$

SE of a difference

$$SE = \sqrt{MSE\left(\frac{1}{n_i} + \frac{1}{n_j}\right)}$$

3 RCBD: one missing value

Substitution estimate

$$\hat{x} = \frac{rB + vT - G}{(r-1)(v-1)}$$

B, T = observed totals of the affected block and treatment; G = observed grand total (all **excluding** the missing value).

Error df

$$(r-1)(v-1) - 1$$

4 Latin square: one missing value

Substitution estimate

$$\hat{x} = \frac{v(R + C + T) - 2G}{(v-1)(v-2)}$$

R, C, T = observed totals of the affected row, column and treatment; G = observed grand total.

Error df

$$(v-1)(v-2) - 1$$

5 Bartlett's covariance method

Create an imaginary **covariate** X: 1 (or −1) for the missing plot, 0 elsewhere. Set $Y = 0$ for the missing plot. Then run **ANCOVA** for the design.

The indicator accounts for the missing position: total and error df are each **one less** than in the complete design.

6 After substituting

- ✓ The estimate **minimises the error SS**.
- ✓ Subtract **one** error df per estimated value.
- ✓ The treatment SS can be **inflated**: apply the bias correction.
- ✓ Comparisons involving the affected treatment need adjusted SEs.
- ✓ Several missing values: estimate by **iteration**.

7 Choosing an approach

Design	Approach	Error df
CRD	Fit observed data (unequal n_i)	$n_{obs} - v$
RCBD	Substitution $\hat{x}$, Bartlett's method, or fit the observed data	$(r-1)(v-1)-1$
Latin square	Substitution $\hat{x}$ or Bartlett's method	$(v-1)(v-2)-1$

! Common mistakes

- × Counting $\hat{x}$ as a real observation.
- × Including the missing cell in B, T or G.
- × Ignoring missingness linked to the response itself (bias).

? Fill in the blanks

1. In an RBD with r = 4 and v = 5, B = 60, T = 50 and G = 400 give a missing value estimate x̂ = ———. [7.5]
2. An RBD with 4 treatments in 5 blocks and one missing plot has ——— error df. [11]
3. A 4 × 4 Latin square with one missing plot has ——— error df. [5]
4. The substituted value is the one that makes the error sum of squares ———. [as small as possible]
5. In a CRD, a lost plot is handled by analysing with ——— replications. [unequal]
6. In Bartlett's method the dummy covariate is ——— for the missing plot and 0 elsewhere. [1]

Source: *Textbook of Agricultural Statistics*, Ch. 23 · Dr. Pratheesh P. Gopinath · Dr. Adarsh V.S. · Mohammed Hisham M. · Dr. Manju Mary Paul · KAU Me-LON

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